Who the Count Missed

The census grades itself, state by state, and mostly returns a blank

84-355 Democracy's Data

August 2026

The census is an attempt to count everyone, and it does not succeed. That much the decennial chapter established, using the national rates by race the Bureau put in its press release: Black residents 3.30% undercounted, Hispanic residents 4.99% undercounted, non-Hispanic White residents 1.64% *over*counted.

This chapter opens the same measurement in the two directions that press release did not. **Where** were people missed — which states, on a map. And **who** was missed, on cuts other than race: by age, by sex, by whether a household owns or rents.

The two answers do not look alike, and the reason they do not is the thing worth carrying out of here.

01 The instrument that measures the instrument

After each census the Bureau runs a second, entirely separate survey called the **Post-Enumeration Survey**. Interviewers go back to a sample of blocks and count the people living there from scratch, without reference to what the census found. Then the two lists are matched person by person. Someone the survey found and the census did not is evidence of an omission; someone the census has twice is evidence of a duplicate. Out of that comparison comes an estimate of the true population, and the gap between it and the published count is **net coverage error**.

For 2020 the survey counted about 10,000 blocks from scratch, roughly 161,000 housing units, and completed about 114,000 household interviews.

Three properties of that design decide everything that follows.

It is a sample, so nothing it produces is a count. The census count of Texas is a count. The estimate of how far off that count was is an estimate, and the Bureau prints a margin next to it saying how firm it considers that estimate to be. Two columns, always together.

It is a net measure. A state where 40,000 people were missed and 40,000 were counted twice comes out at zero, the same as a state where nothing went wrong. Net coverage error near zero is not a certificate of accuracy; it is a statement about the *balance* of two errors, not their size.

It covers the household population only. People in dormitories, prisons, nursing homes and shelters are enumerated by a different operation and evaluated separately, and the Remote Alaska areas are outside it altogether. The survey's national base is 323,200,000 people against a census resident population of 331,449,281 — about **8,249,281 people who are in no figure in this chapter.**

The national line reads as an undercount of **0.24%**, about 777,546 people, with a margin of 0.25 beside it and **no mark**. The Bureau publishes the number and declines to say the country was undercounted. **That combination, a number plus a refusal to stand behind it, is the shape of most of this chapter.**

What follows is that same survey, cut finer.

Check	Value
state rows parsed out of Appendix Table 3	50 states and DC, out of 52 printed rows
state names in alphabetical order, matched to FIPS	51 matched, 0 unmatched
the Bureau's marks, checked against the numbers printed beside them	50 of 51 track; Delaware is the exception
national estimate, identical in all four PES tables	-0.24% in all four
self-response rate, national, against the published figure	67.0%, as published
internet self-response below overall, every state	51 of 51
map rings joined to a coverage estimate	128 of 128 rings
median published margin, one state against one national group	1.43 against 0.48 percentage points

> **Before you read on, commit to a number.** The Bureau publishes a net coverage error for each of the 50 states and the District of Columbia — 51 numbers — and marks the rows where it is willing to say which way the count went. **On how many of the 51 do you think it puts that mark?** Write the number down before you turn the page.

02 Fifty-one numbers

Here is the whole state table as a distribution, before any of it is mapped.

Statistic	Value
Estimates published	51
Mean	+0.18%
Median	+0.10%
Standard deviation	2.63 points
Range	-5.04% (Arkansas) to +6.79% (Hawaii)
Interquartile range	-1.27% to +1.51%
Rows the Bureau marks	14 of 51 — 6 undercounts, 8 overcounts
Margin printed beside it, narrowest	0.65 points (California)
Margin printed beside it, widest	4.93 points (District of Columbia)
Margin printed beside it, median	1.43 points

Read the middle rows first. The estimates are centred essentially on zero, with a mean of +0.18% and a median of +0.10%. They scatter widely around it, from -5.04% in Arkansas to +6.79% in Hawaii. A 2.63-point standard deviation across states, around a national figure of 0.24%, is the first sign that the country's near-perfect total is an average over parts that are not.

Now read the last three rows, because they are what the middle rows are made of. **The margin the Bureau prints beside a state's number is usually wider than the number itself.** The median margin is 1.43 points; the median estimate is 0.10. When a publisher hedges a number by more than the number is worth, it is telling you something, and here it tells you exactly how far the table can be pushed.

So the Bureau marks 14 rows of 51. **6 states it calls undercounted**: Arkansas, Florida, Illinois, Mississippi, Tennessee, and Texas. **8 it calls overcounted**: Delaware, Hawaii, Massachusetts, Minnesota, New York, Ohio, Rhode Island, and Utah. On the remaining **37**, which is every other state plus the District of Columbia, it publishes a number and says nothing about which way the count went.

This is the whole reading problem of the chapter, and it is not a problem about statistics. It is the ordinary problem of this book: **the source shipped a caveat attached to every number, and the caveat is the part that falls off first.** A table with 51 rows travels; a table with 51 rows, 51 margins and 14 marks does not fit in a sentence. What follows is what the full table looks like when none of it is dropped.

03 The map

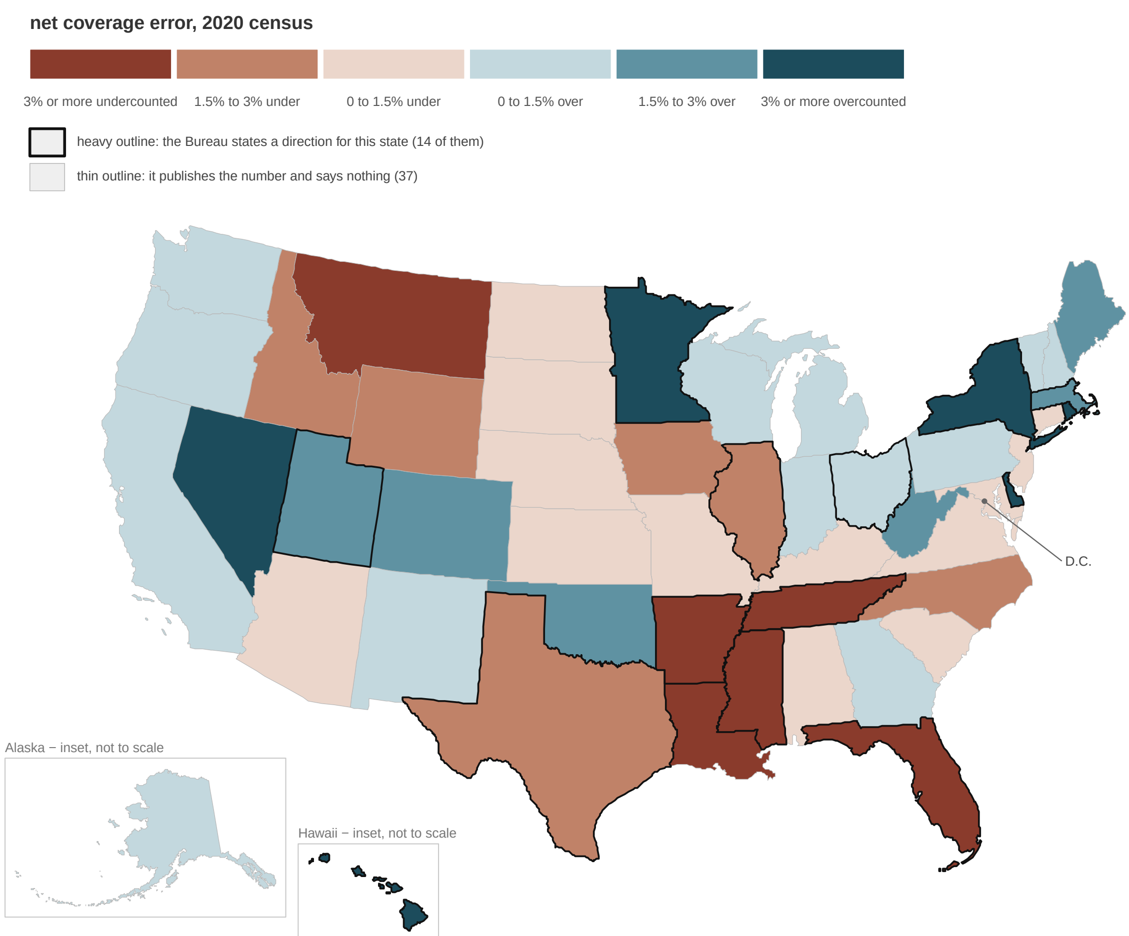

Figure 1. Net coverage error in the 2020 census, by state. Fill is the number the Bureau published: rust where the count is estimated to have missed people, slate where

it is estimated to hold more people than live there. The outline is the Bureau's own mark — whether it will say, for that state, which way the count went. Only 14 states carry the heavy outline.

Two channels, because the source has two columns and dropping either one produces a different map. Color only the 14 marked states and 37 published numbers vanish from the record. Color all 51 without the outlines and the map claims more than the Bureau did. It invites you to see a rust-colored Montana and conclude something about Montana that the agency which measured Montana would not put its name to.

Look for a pattern in the fill and you will half-find one. The deep rust runs along the Gulf and the lower Mississippi; the deep slate sits in New England, the upper Midwest and Hawaii. Now look at the outlines, and watch most of that pattern turn out to be shading the Bureau printed without endorsing. District of Columbia posts +4.59%, the largest number on the map that carries no mark, and it is thin-outlined. So are Montana, Nevada and Louisiana, each of them past three points.

A thin outline does not mean the state was counted well. It means this survey did not see enough of that state to say. Those are different claims, and the map is drawn the way it is so that they stay different.

04 The bigger number is not the safer one

Put five states side by side, exactly as the source prints them.

State	Estimate	Margin printed beside it	The Bureau says
Illinois	-1.97%	0.89	undercounted
Louisiana	-3.73%	2.36	nothing
Ohio	+1.49%	0.67	overcounted
District of Columbia	+4.59%	4.93	nothing
Montana	-4.39%	3.70	nothing

Illinois's number says the count missed 1.97% of its people. Louisiana's says 3.73% — nearly twice as much. **The Bureau calls Illinois undercounted and says nothing about Louisiana.**

That is not a claim that Louisiana was counted better. Look at the middle column. Illinois's margin is 0.89 points and Louisiana's is 2.36: the survey looked at Illinois closely enough to describe it and did not look at Louisiana closely enough to describe it. Same table, same census, different amount of evidence underneath.

The smallest number on the page the Bureau *will* characterise is Ohio's +1.49%, carrying a margin of 0.67. The largest it will not is District of Columbia's +4.59%, carrying a margin of 4.93. **The mark tracks the second column, not the first.**

Where do the margins come from? Mostly from how much of each state the survey saw. California, the largest state, has the narrowest margin on the map at 0.65 points; District of Columbia has the widest at 4.93. **The width of a margin is a fact about the survey, not about how well a state was counted** — and it is what decides whether the state shows up in anybody's headline.

So the practical rule for using this table is short, and it needs no arithmetic: **read across the row.** A number from Appendix Table 3 quoted without its margin and its mark is not the Bureau's claim. It is a number that was standing next to the Bureau's claim.

What the rates are in people

Percentages are easy to nod at, so here are three of them as bodies.

State	Census count	Rate	Implied
Texas	28,540,000	-1.92%	558,695 missed
Florida	21,070,000	-3.48%	759,673 missed
New York	19,590,000	+3.44%	651,485 in excess

The arithmetic is not the one you would reach for first. A coverage rate is measured against the *true* population rather than the published one, so the implied gap is the published count divided by one plus the rate, minus the published count. Texas's 1.92% works out to 558,695 people rather than the 547,968 that taking the rate straight off the published count would give.

Hold those next to a House seat. The 2020 apportionment gave one seat per 761,169 people, so Florida's implied gap is 99.8% of a seat, Texas's 73.4%, and New York's excess 85.6%. Not one of the 51 clears a whole seat outright, and Florida comes within a fraction of one — but a whole seat is not the bar these have to clear. The last seat in the 2020 apportionment turned on a margin of a few dozen people, which is the subject of the apportionment chapter. **The rates in Figure 1 are not a rounding concern.**

Adding the 51 implied figures together would not give a national total. 37 of them rest on numbers the Bureau itself will not characterise, and adding those to the 14 it will would produce a total nobody published and nobody could defend. The national figure of 777,546 comes from the national rate applied to the national count, which is a different calculation and one the source supports.

05 Who answered on their own, which is a different question

Alongside coverage error the Bureau publishes something that looks related and is not. It is the **self-response rate**: the share of housing units that filled the form in by internet, phone or mail, before anyone was sent to knock. Nationally 67.0% did, with 53.5 points of that online. It is the first census where most of the country answered over the internet.

That number is a *count*, not an estimate. Every housing unit either responded or it did not, the Bureau knows which, and this column arrives with no margin beside it and no mark on it — there is nothing to hedge. By state it runs from 54.7% in Alaska to 75.1% in Minnesota.

It is tempting to treat it as a proxy for accuracy — states that answered less were counted worse. Test it.

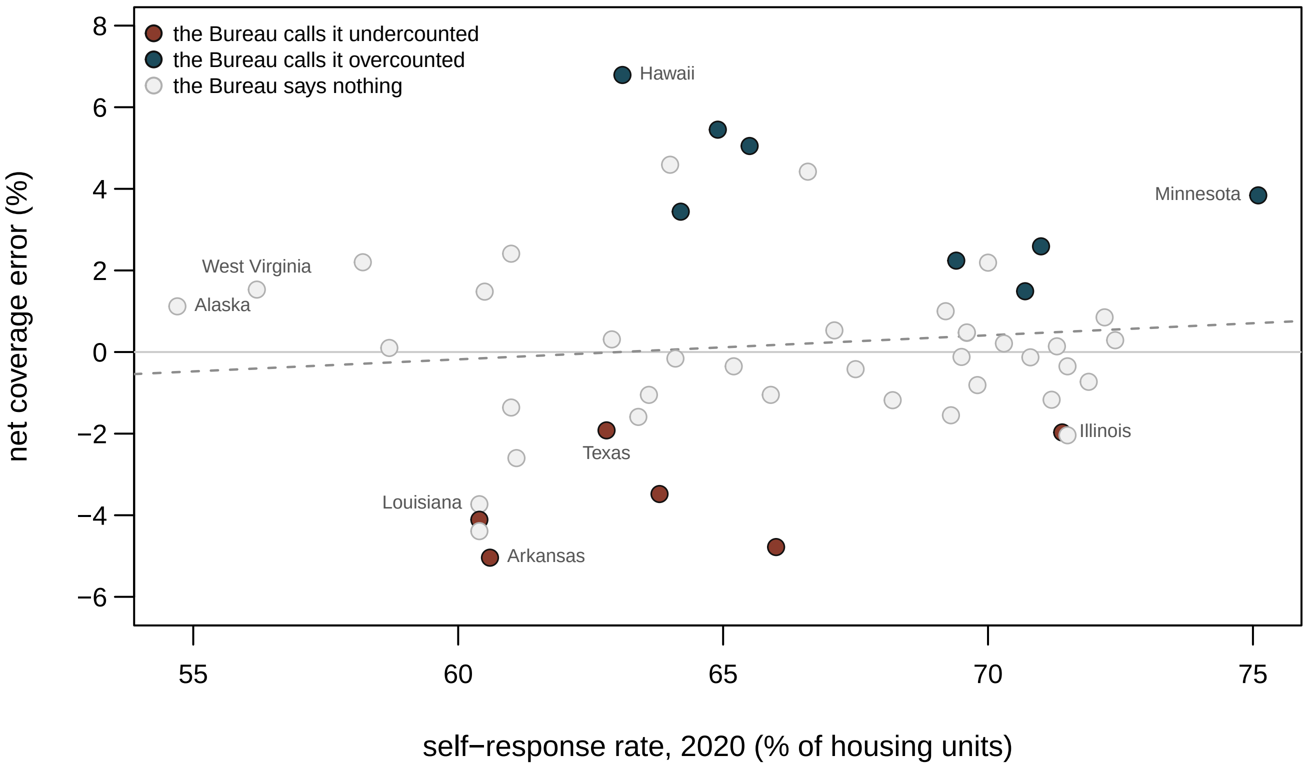

Figure 2. Self-response against net coverage error, by state. Each circle is a state; the dashed line is the least-squares fit. The correlation is **0.11**.

0.11 is nothing. Alaska had the lowest self-response in the country at 54.7% and came out +1.12%, an overcount. Minnesota had the highest at 75.1% and came out +3.84%, also an overcount, and one the Bureau states. Illinois answered at 71.4%, well above the national rate, and is one of the six states the Bureau calls undercounted.

The reason is not mysterious once the operation is in view. A household that does not answer is not thereby uncounted — it is *followed up*, in person, by a census taker, and in 2020 that operation reached 33.0% of housing units. Self-response measures how much of the count came in by mail and internet rather than by knocking. **Nonresponse is a cost, not an omission.**

There is a relationship in this data, but it is with the other column.

Comparison	Correlation	Reading
Self-response against the coverage estimate	0.11	no relationship worth naming
Self-response against that estimate's standard error	-0.47	states that answered less got a wider margin

Self-response is 0.47 correlated with the *margin* — states where fewer households answered are states the Bureau hedged harder about. That is a statement about the evaluation rather than about the count. Harder-to-reach populations are harder to reach for the follow-up operation, and harder to reach for the survey that grades it. So the places where you would most want a sharp estimate are the places where the estimate is blurriest. It is also only a correlation across 51 points, and states differ in a hundred other ways at the same time.

Two numbers about the same state, both published by the same agency, that answer different questions. One is an exact count of how the form came back. The other is a sampled estimate of who is missing. Neither substitutes for the other, and the temptation to let the precise one stand in for the uncertain one is exactly the move to resist.

06 Where the signal is

The same survey, cut by demographic group instead of by state, stops being blank. Here is age.

Age	Net coverage error	Margin	The Bureau says
Total	-0.24%	0.25	nothing
0 to 17	-0.84%	0.38	undercounted
0 to 9	-1.40%	0.49	undercounted
0 to 4	-2.79%	0.64	undercounted
5 to 9	-0.10%	0.56	nothing
10 to 17	-0.21%	0.43	nothing
18 to 29	-1.62%	0.57	undercounted
30 to 49	-1.47%	0.35	undercounted
50 and over	+1.65%	0.25	overcounted

And age crossed with sex, which is where it gets sharp.

Group	Net coverage error	Margin	The Bureau says
18 to 29 males	-2.25%	0.57	undercounted
18 to 29 females	-0.98%	0.58	undercounted
30 to 49 males	-3.05%	0.35	undercounted
30 to 49 females	+0.10%	0.36	nothing
50-and-over males	+0.55%	0.25	overcounted
50-and-over females	+2.63%	0.25	overcounted

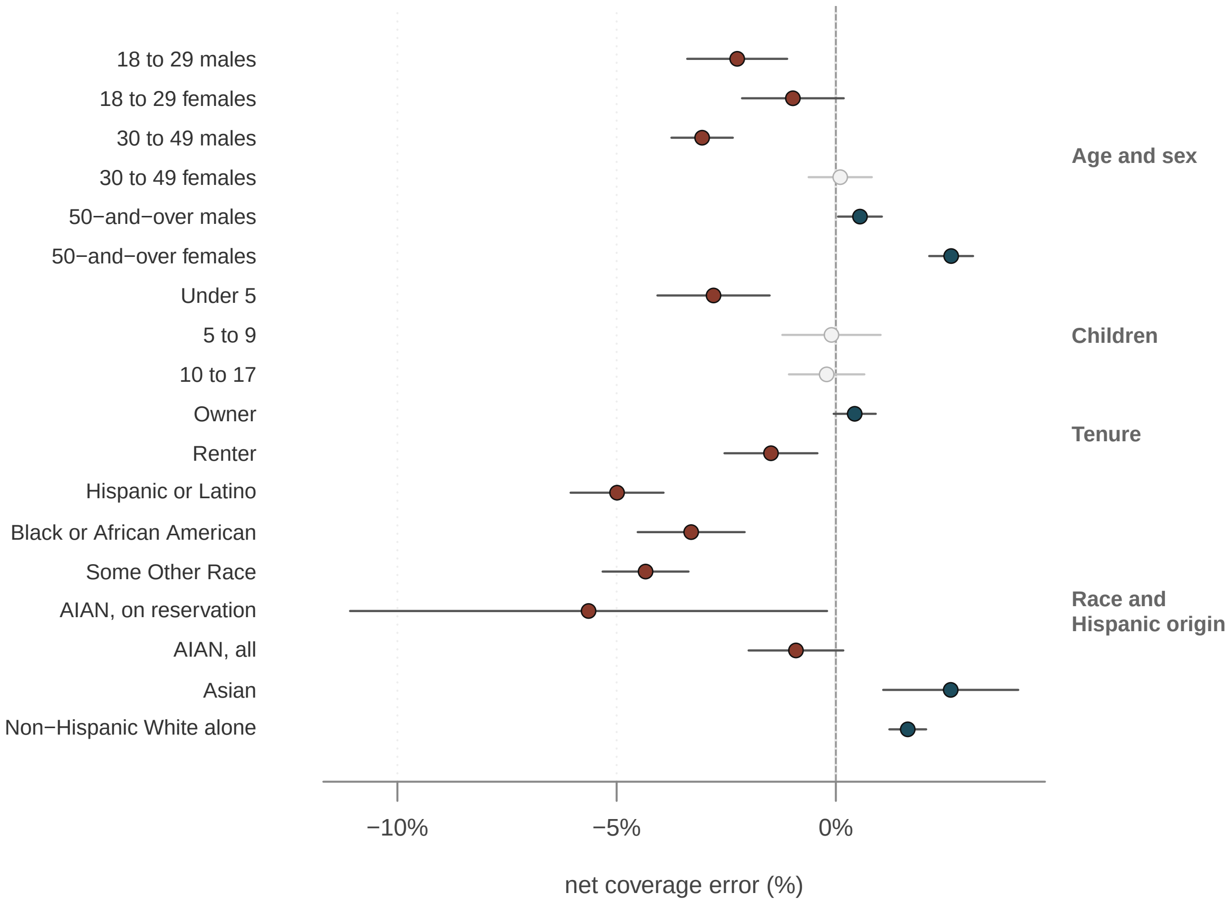

Figure 3. Every national demographic number in this chapter, with the margin the Bureau prints beside it. Filled circles are the rows the Bureau characterises; hollow circles are the rows where it publishes a number and says nothing. Rust is an undercount, slate an overcount. The bars are the published margin drawn to scale, and the thing to notice is how short they are: a median of 0.48 points here against 1.43 for a state on Figure 1. **This is the same survey, hedged far less.**

Four things in that figure are worth saying out loud.

Children under five are undercounted by 2.79% — one of the largest rates in the whole survey, and one of the most persistent across censuses. Older children are not: 5-to-9-year-olds come in at -0.10% and 10-to-17-year-olds at -0.21%, and the Bureau characterises neither. Whatever goes wrong is going wrong on the form, in households, about the youngest people in them.

Age and sex together separate what either alone would hide. Men aged 30 to 49 are undercounted 3.05%. Women in the same age band come in at +0.10%, which the Bureau declines to characterise. Men aged 18 to 29, -2.25%; women, -0.98%. Read only the age groups and you would see a modest -1.47% for everyone in their thirties and forties, which is the average of a large undercount and nothing.

Women aged 50 and over are overcounted 2.63%, the largest overcount in the demographic tables. An overcount is not a harmless error in the other direction: it is duplicates and people counted in the wrong place, and it inflates a population just as surely as an omission deflates it.

Renters are undercounted 1.48% and owners overcounted 0.43%. That gap has been in every one of the last four censuses.

Tenure	2020	2010	2000	1990
Owner	+0.43%	+0.57%	+1.25%	-0.04% (ns)
Renter	-1.48%	-1.09%	-1.14%	-4.51%

And race, the cut the decennial chapter opened, in the Bureau's full detail rather than the six rows of the press release:

Group	Net 2020	Margin	The Bureau says	Net 2010
Total	-0.24%	0.25	nothing	+0.01%
White	+0.66%	0.21	overcounted	+0.54%
Non-Hispanic White alone	+1.64%	0.21	overcounted	+0.83%
Black or African American	-3.30%	0.61	undercounted	-2.06%
Asian	+2.62%	0.77	overcounted	0.00%
American Indian or Alaska Native	-0.91%	0.54	undercounted	-0.15%
On Reservation	-5.64%	2.72	undercounted	-4.88%
American Indian Areas Off Reservation	+3.06%	2.72	nothing	+3.86%
Balance of the United States	-0.86%	0.47	undercounted	+0.05%
Native Hawaiian or Other Pacific Islander	+1.28%	2.11	nothing	-1.02%
Some Other Race	-4.34%	0.49	undercounted	-1.63%
Hispanic or Latino	-4.99%	0.53	undercounted	-1.54%

These rows are **alone or in combination**: someone who marked two races appears in two of them, so the column does not add to anything and should not be made to. The two indented rows under American Indian or Alaska Native are parts of it. They are the 5.64% undercount on reservations, against 0.86% for American Indian and Alaska Native residents living elsewhere. The Bureau averaged those two into the 0.91% above them. That is why the headline rate for the group is smaller than the rate in the places where the problem is worst.

07 Why the map and the tables do not reconcile

You now have two pictures of one survey. The demographic tables are sharp: large rates, short margins, and the Bureau willing to characterise nearly every row. The map is mostly blank: 37 of 51 states where it publishes a number and stops there.

They are not in tension, and the reason is about how much evidence sits under each row rather than about the country.

A national group is measured with the whole sample; a state is measured with one state's slice of it. Every one of the 114,000 interviews contributes to the estimate for renters. Only the interviews conducted in Wyoming contribute to the estimate for Wyoming. That is the entire difference, and it shows up as the width of the interval — a median of 0.48 points against 1.43.

A state's number is a mixture, and mixtures cancel. A state is not undercounted or overcounted as such; the *people* in it are, at rates that differ by age, sex, tenure and race. A state with many renters and many young men and many older women is pulling in both directions at once, and its net figure is what is left after the pulls have partly cancelled. Hawaii's +6.79% and Arkansas's -5.04% are the residue of that mixing, not a property of the place.

So the demographic tables are the mechanism and the map is the consequence, imperfectly observed. If you want to know who the census misses, read Figure 3. If you want to know what that did to a particular state's count, the honest answer for 37 states is the one the Bureau itself gives. Here is a number, and we are not going to tell you which way it points.

08 What the survey is allowed to do

Nothing. The Post-Enumeration Survey cannot correct the census, and it was never going to be able to.

The rule is narrow and it is the important one. Congress told the Bureau it may use sampling in its work "except for the determination of population for purposes of apportionment". In 1999 the Supreme Court read that section to bar a sampled apportionment count. The survey in this chapter is a sample. The counts that hand out House seats therefore have to be the counts that were taken, whatever this survey says about them.

Nor is the Bureau obliged to adjust anything else, and that was settled on this instrument. After the 1990 census the Bureau ran a post-enumeration survey, found an undercount, and the Secretary of Commerce declined to use it to correct the published figures. New York City and others sued, because an uncorrected count cost them representation and money. In *Wisconsin v. City of New York* (1996) the Court upheld the refusal as a reasonable exercise of the discretion Congress gave the Secretary.

So this survey exists to describe a problem, not to repair it. It is a rare kind of official document: an agency's careful measurement of how wrong its own flagship product is, published in full, with no mechanism attached for acting on it. The decennial chapter sets out the statute and the three cases.

That is worth holding next to the two refusals this chapter has already shown. The Bureau declines to characterise 37 states because the survey did not see enough of them. It declines to correct any state because the law does not permit it. **The first is a limit of evidence and the second is a limit of authority**, and they are easy to mistake for each other on a page where both produce the same blank.

09 What this chapter cannot tell you

Net coverage error is not accuracy, and the Bureau's own components table is the proof. Every headline number in this chapter is a *balance*; here is what the national balance is made of.

Component	Percent	Standard error
Correct enumerations	94.40%	—
Erroneous: duplication	1.60%	—
Erroneous: other reasons	0.60%	—
Whole-person census imputations	3.40%	—
Correct, dual-system estimate	94.20%	0.20
Omissions	5.80%	0.20
Net coverage error	-0.24%	0.25

About 5.8% of people were omissions and about 3.4% of the published count were whole-person imputations — a record the Bureau filled in because nothing usable came back for that person. Both are an order of magnitude larger than the 0.24% net they produce. **A state at zero is not a state where nothing happened**; it is a state where a great deal happened in both directions and the two sides came out close to even.

No state-by-race table exists, and it is the table everybody wants. The Bureau does not publish coverage by state crossed with race, age or tenure. The decennial chapter illustrates the scale by putting national rates on one state's counts and says plainly that this is an illustration rather than an estimate. Nothing here improves on that. **You cannot get from Figure 3 to a claim about who was missed in Georgia.**

A blank is not a clean bill of health. On 37 states the Bureau publishes a number and declines to characterise it. That is the agency saying *this survey did not see enough of this state to tell you*, and it gets reported as *this state was fine* constantly. The correct reading of a thin outline on Figure 1 is **we do not know**, and staring harder at the number will not change it — the missing evidence is missing.

The 2020 and 2010 margins are not the same quantity, so the two columns do not subtract. The state table labels the 2020 margin a standard error and the 2010 one a root mean squared error. The 2010 estimates carried a bias that the 2020 modeling removed, and the Bureau widened the older margin to cover it. Subtraction will hand you a change for every state, and the Bureau marked none of them, so this chapter reports none. **"It got worse since 2010" is a claim the source does not support at the state level.**

Puerto Rico is absent from the map, because the state table does not evaluate it. The Post-Enumeration Survey covered Puerto Rico on a separate design and published it separately, and folding an estimate from one design into a map of another would have been the wrong kind of completeness.

Three numbers here were keyed in rather than read. The 10,000 blocks, 161,000 housing units and 114,000 interviews are the survey's design, which the Bureau states in prose rather than publishing as data; they were transcribed. Every estimate, every margin and every one of the Bureau's marks was taken out of the reports mechanically instead, so nothing that carries an argument depends on somebody having typed it correctly.

Secondary summaries of this survey are not always right, and one worth naming got a sign backwards. The National Conference of State Legislatures' *Redistricting Law 2020* summarises the 2010 results for legislators and their staff. Most of its figures match the Bureau's: Black residents undercounted by about 2.06%, Hispanic residents by about 1.54%, residents on reservations by about 4.88%. But it reports the non-Hispanic White population as **undercounted** by 0.8% in 2010, where the Bureau's own table gives an *over-count* of 0.83%. The direction is the whole point of that row, and a reader taking the summary on trust would have it exactly reversed. Read the agency's table, not a description of it.

The estimates are frozen at the 2022 releases. These are the March 2022 demographic report and the May 2022 state report as published. The Bureau has revised a PES table before. A note in the state report records that Figure 2 and one appendix table were revised after first release, and a later revision would not be noticed here.

A count is not made true by being measured. The Post-Enumeration Survey is itself a survey, with its own nonresponse, its own matching errors, and its own model. It is the best available evidence about the census and it is evidence of the same general kind as the thing it evaluates. The Bureau's own accuracy report is candid about this; nothing in a coverage estimate is a fact in the way a count of returned forms is a fact.

10 Sources

Data

- U.S. Census Bureau. *Census Coverage Estimates for People in the United States by State and Census Operations*, 2020 Post-Enumeration Survey Estimation Report, May 2022 release. Appendix Table 3 is the state table used throughout: census count, 2020 estimate, the margin beside it, and the 2010 estimate with its own margin. The asterisks in that table are the Bureau's marks — its footnote calls them estimates "significantly different from zero", which is the agency's way of saying *this is a row we will characterise* — and they are what the outlines in Figure 1 draw. https://www2.census.gov/programs-surveys/decennial/coverage-measurement/pes/census-coverage-estimates-for-people-in-the-united-states-by-state-and-census-operations.pdf
- U.S. Census Bureau. *National Census Coverage Estimates for People in the United States by Demographic Characteristics*, 2020 Post-Enumeration Survey Estimation Report, March 2022 release. Table 4 is race and Hispanic origin, Table 6 tenure across four censuses, Table 10 age crossed with sex, Table 11 age alone. https://www2.census.gov/programs-surveys/decennial/coverage-measurement/pes/national-census-coverage-estimates-by-demographic-characteristics.pdf
- Both of those are **PDFs, and there is no data release behind them** — no CSV, no spreadsheet, no API endpoint anywhere on the Bureau's site carries these numbers. Anyone who wants them in a table has to take them out of the text layer of a printed report, which is what was done here for all 51 states and all four demographic tables. Two details are worth knowing if you go and do it yourself: the reports set minus as an en dash, so every undercount fails to parse as a number until that is undone, and the indentation of a row label is the only thing recording that "On Reservation" is a subdivision of the row above it rather than a category beside it.
- U.S. Census Bureau. *2020 Census self-response rates*, the file behind the Bureau's response-rate map, `decennialrr2020.csv` — every geography, one row per posting date, no key required. https://www2.census.gov/programs-surveys/decennial/2020/data/tracking-response-rates/self-response-rates-map/ A trap in that file is worth naming, because its column names invite the mistake: `CRRALL` is the geography's own cumulative self-response rate, while `CMAX` is the *maximum over the geographies inside it*. Alabama's `CRRALL` is 63.6 and its `CMAX` is 77.8; the second number belongs to some tract in Alabama, not to Alabama. The rates above are `CRRALL` at the final post-

ing, 2021-01-29, and the national figure comes back at 67.0% exactly as published.
- State outlines are the Census cartographic-boundary state file, projected and simplified for the migration chapter and reused here unchanged rather than re-derived — a second projection of the same boundaries would only drift out of agreement with the first.

The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

`age.csv`, `age_sex.csv`, `components.csv`, `map_bins.csv`, `map_insets.csv`, `map_states.csv`, `race.csv`, `states.csv`, `tenure.csv`

Your turn

None of these is answered above, and every one can be answered from the tables you just opened.

- `states.csv` carries `est_2020` beside `se_2020`, and `bureau_states_it` records whether the Bureau was willing to say the number out loud. Count the TRUE rows. What share of the country got an answer it was prepared to stand behind?
- `states.csv` also has `self_response` and `internet_response`. Sort by `self_response` and set it against `est_2020`. Do the states that answered on their own get counted better?
- `race.csv` puts the 2020 estimate beside the 2010 one for each group. Which groups moved, and in which direction? Then ask whether that is the count improving or the measure changing.
- `components.csv` breaks the count into correct enumerations and everything else. Work out what share is not a correct enumeration, and decide whether that number surprises you.

Statute and cases

- Census Act, 13 U.S.C. § 195 (sampling, "except for the determination of population for purposes of apportionment").
- *Wisconsin v. City of New York*, 517 U.S. 1 (1996) — the Secretary's refusal to adjust the 1990 count using that decade's post-enumeration survey, upheld.
- *Department of Commerce v. U.S. House of Representatives*, 525 U.S. 316 (1999) — sampling barred for the apportionment count.
- National Conference of State Legislatures, *Redistricting Law 2020* (Denver: NCSL, October 2019), ch. 1, whose sidebar summarises the 2010 coverage results for legislative staff. Cited above as a worked example of a secondary summary disagreeing with its own source.

Related

- U.S. Census Bureau. "2020 Census Undercounts in Six States, Overcounts in Eight," *America Counts*, 19 May 2022 — the Bureau's own summary of the state table. https://www.census.gov/library/stories/2022/05/2020-census-undercount-overcount-rates-by-state.html
- U.S. Census Bureau. *Source and Accuracy of the 2020 Post-Enumeration Survey Estimates* — what the survey itself can and cannot do.
- The decennial census chapter for the instrument being evaluated, and the apportionment chapter for what a few hundred thousand people are worth in seats.

What the build verified before writing any of the above

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state rows parsed out of Appendix Table 3	50 states and DC, out of 52 printed rows
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the Bureau's marks, checked against the numbers printed beside them	50 of 51 track; Delaware is the exception
national estimate, identical in all four PES tables	-0.24% in all four
self-response rate, national, against the published figure	67.0%, as published
internet self-response below overall, every state	51 of 51
map rings joined to a coverage estimate	128 of 128 rings
median published margin, one state against one national group	1.43 against 0.48 percentage points

REBUILD THIS DATA YOURSELF, WITH AN AI ASSISTANT

```
I want to rebuild a public dataset from its original sources, without
any starting files. Please do the following and show your work.
THE SOURCES. U.S. Census Bureau, three of them, all open, no key.
1. "Census Coverage Estimates for People in the United States by
State and Census Operations", Post-Enumeration Survey report
PES20-G-02, May 2022:
https://www2.census.gov/programs-surveys/decennial/coverage-
measurement/pes/census-coverage-estimates-for-people-in-the-united-
states-by-state-and-census-operations.pdf
Appendix Table 3 is the state table: for each state, the census
count, the estimated net coverage error, its standard error, and
an asterisk wherever the Bureau stands behind a direction.
2. "National Census Coverage Estimates for People in the United
States by Demographic Characteristics", report PES20-G-01, March
2022:
https://www2.census.gov/programs-surveys/decennial/coverage-
measurement/pes/national-census-coverage-estimates-by-demographic-
characteristics.pdf
National estimates by race and Hispanic origin, by age and sex,
and by whether the home is owned or rented.
3. The 2020 Census self-response rates, one CSV of about 7.6 MB:
https://www2.census.gov/programs-surveys/decennial/2020/data/tracking-
response-rates/self-response-rates-map/decennialrr2020.csv
```

One row per geography, from the nation down to tracts.

HOW TO READ THEM. The two reports exist only as PDF — no CSV, no API, no spreadsheet anywhere on the Bureau’s site. Extract the text with a layout-preserving tool (poppler’s pdftotext with its -layout flag works) and parse the tables out of the text. Two traps. In the PDFs, negative numbers are printed with an en dash rather than a hyphen-minus; convert it or every undercount refuses to be a number. In the rates CSV, the column you want is CRRALL, the cumulative self-response rate. The columns CMAX and CINTMAX read like final rates and are nothing of the sort: they are the maximum over the geographies inside each row’s area. Alabama’s CRRALL is 63.6; its CMAX is 77.8, and the second number is some tract in Alabama, not Alabama.

WHAT TO BUILD.

1. The state table: 50 states and DC, each with its net coverage estimate, its standard error, and whether the Bureau marked it — and a count of the marked ones, split undercount against overcount.
2. The national estimates by demographic group, from the second report, with their standard errors.
3. Self-response by state from CRRALL, plus the national rate.

CHECK YOUR WORK. The copies this book used were fetched in August 2026. If your rebuild matches them, you should find:

- a national net coverage estimate of –0.24%, printed identically in all four tables of the demographic report
- the Bureau marks 14 of the 51 state estimates: six undercounts and eight overcounts
- state standard errors run from 0.65 to 4.93 percentage points
- the rates file has 123,248 rows, and the national self-response rate is 67.0%

Everything here is final: the PES reports will not be revised, the rates file stopped moving in 2021, and the next survey of this kind follows the 2030 census.